

assignment_circular motion bonus A↓

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:55pm

Quiz Instructions

2 attempts

There is a tutorial



Question 1 7.2 pts

When an object moves in uniform circular motion, the direction of its acceleration is

☐

is directed toward the center of its circular path.

☐

depends on the speed of the object.

☐

in the same direction as its velocity vector.

☐

in the opposite direction of its velocity vector.

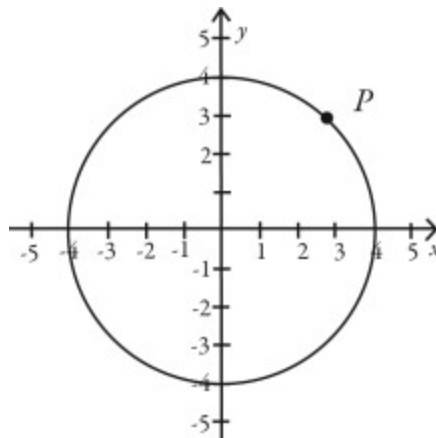
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is directed away from the center of its circular path.



Question 2 7.2 pts

Point P in the figure indicates the position of an object traveling at constant speed clockwise around the circle. Which arrow best represents the direction of the acceleration of the object at point P ?





Question 3 7.2 pts

An aircraft performs a maneuver called an "aileron roll." During this maneuver, the plane turns like a screw as it maintains a straight flight path, which sets the wings in circular motion. If it takes it 35 s to complete the circle and the wingspan of the plane is 11 m, what is the acceleration of the wing tip?

- ☐ 0.18 m/s²
- ☐ 0.99 m/s²
- ☐ 5.6 m/s²
- ☐ 1.0 m/s²



Question 4 7.2 pts

A ball is tied to the end of a cable of negligible mass. The ball is spun in a circle with a radius 2.00 m making 7.00 revolutions every 10.0 seconds. What is the magnitude of the acceleration of the ball?

- ☐ 29.3 m/s²
- ☐ 67.9 m/s²
- ☐ 74.2 m/s²

38.7 m/s²14.8 m/s²

Question 5 7.2 pts

An electrical motor spins at a constant 2857.0 rev/min. If the armature radius is 2.685 cm, what is the acceleration of the outer edge of the armature?

844.4 m/s²84.40 m/s²241,100 m/s²2403 m/s²

Question 6 7.2 pts

You are making a circular turn in your car on a horizontal road when you hit a big patch of ice, causing the force of friction between the tires and the road to become zero. While the car is on the ice, it



continues to follow a circular path, but with a radius larger than the original radius.



moves along a straight-line path toward the center of the circle.



moves along a path that is neither straight nor circular.



moves along a straight-line path in its original direction.



moves along a straight-line path away from the center of the circle.



Question 7 7.2 pts

When a car goes around a circular curve on a horizontal road at constant speed, what force causes it to follow the circular path?



gravity



No force causes the car to do this because the car is traveling at constant speed and therefore has no acceleration.



the normal force from the road



the friction force from the road



Question 8 7.2 pts

For an object in uniform circular motion, its velocity and acceleration vectors are always perpendicular to each other at every point in the path. (slide 57)



True



False



Question 9 7.2 pts

If an object travels at a constant speed in a circular path, the acceleration of the object is (ac is proportional to speed squared and inversely proportional to the radius)



smaller in magnitude the smaller the radius of the circle.



in the same direction as the velocity of the object.



in the opposite direction of the velocity of the object.



zero.



larger in magnitude the smaller the radius of the circle.



Question 10 7.2 pts

A ball is tied to the end of a cable of negligible mass. The ball is spun in a circle with a radius 2.00 m making 7.00 revolutions every 10.0 seconds. What is the magnitude of the acceleration of the ball?

hint: See lecture 9_27_22 slide 56



38.7 m/s²



74.2 m/s²



29.3 m/s²



14.8 m/s²



67.9 m/s²



Question 11 7.2 pts

You are making a circular turn in your car on a horizontal road when you hit a big patch of ice, causing the force of friction between the tires and the road to become zero. While the car is on the ice, it

- ☐ moves along a path that is neither straight nor circular.
- ☐ moves along a straight-line path away from the center of the circle.
- ☐ continues to follow a circular path, but with a radius larger than the original radius.
- ☐ moves along a straight-line path toward the center of the circle.
- ☐ moves along a straight-line path in its original direction.



Question 12 7.2 pts

A car is moving along a circle of radius R and its speed is V . If the car increases its speed by a factor of 3 ($\times 3$), the centripetal acceleration (or force) is (suppose R stays the same)

hint: you can suppose $V=1\text{m/s}$, $R = 1\text{m}$. Find a_1 . Then take $V = 3\text{m/s}$, $R = 1\text{m}$ find a_2 . What is a_2/a_1

- ☐ $\times 9$
- ☐ $\times 3$
- ☐ $\times 1$
- ☐ stays the same



Question 13 7.2 pts

A car is moving along a circle of radius R and its speed is V . If the car decreases its speed by a factor of 2 (divided by 2), the centripetal acceleration (or force) is (suppose R stays the same)

hint: you can suppose $V=4\text{m/s}$, $R = 1\text{m}$. Find a . Then take $V = 2\text{m/s}$, $R = 1\text{m}$ find a

- ☐ stays the same
- ☐ divided by 4
- ☐ divided by 6
- ☐ divided by 2



Question 14 7.2 pts

A car is moving along a circle of radius R and its speed is V . If for the same speed, the radius is divided by 3 then the acceleration

hint: you can suppose $V = 1\text{m/s}$, $R = 6\text{m}$ find a . Then.. you get it.

☐

is divided by 3

☐

is multiplied by 9

☐

is multiplied by 3

☐

divided by 6



Question 15 5 pts

Question 9: If a vector is entirely in the third quadrant and has an x-component of -2 and a y-component of -2, what is the magnitude of this vector?

A) $2\sqrt{2}$

B) 4

C) $4\sqrt{2}$

D) 8

☐

A

☐

B

☐

D

☐

C



Question 16 5 pts

Question 6: If a vector in the third quadrant has an x-component of -8 and a y-component of -15, what is the angle it makes with the negative x-axis?

hint: Trace the vector. You can use the App <https://phet.colorado.edu/en/simulations/vector-addition>
 [_ \(https://phet.colorado.edu/en/simulations/vector-addition\)](https://phet.colorado.edu/en/simulations/vector-addition)

Notice that I am asking for the angle with the NEGATIVE x-axis (not positive)

☐

242

☐

28

☐

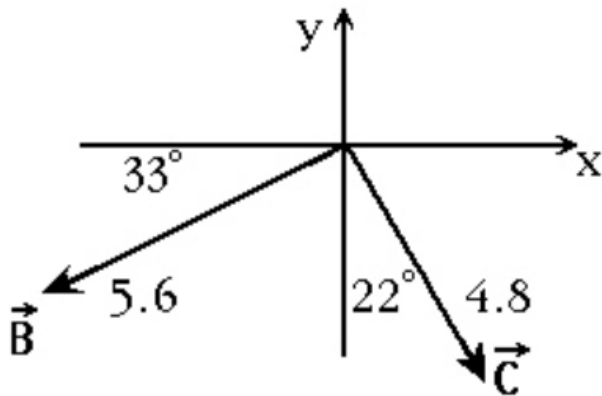
62

☐

118

☒

Question 17 4 pts



The angle of B relative to the x-axis is (going CCW)

☐

147

☐

33

☐

123

☐

213

☒

Question 18 4 pts

Building on previous question. The components of B are

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(3, 4.7)

☐

(-3, -4.7)

☐

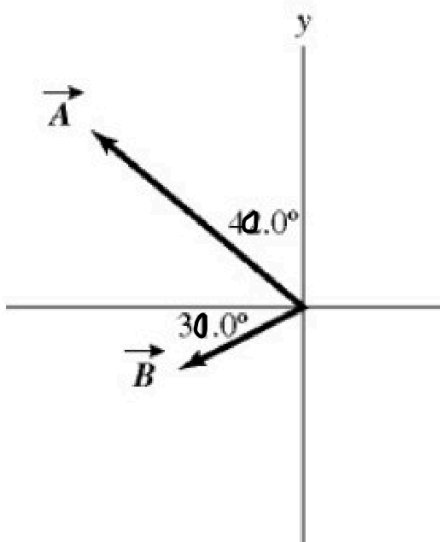
(-4.7, -3)

☐

(4.7, -3)

☒

Question 19 4 pts



A has a magnitude of 10lbs (40 degrees E of N) B has a magnitude of 3lbs (30 degrees S of E)

What are the components of A and B ? (use your polar graph if necessary).

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A(6.43, 7.7) and B(-2.6, -1.5)

☐

A(-7.7, 6.43) and B(-2.6, -1.5)

☐

A(-6.43, 7.7) and B(-2.6, -1.5)

☐

A(6.43, 7.7) and B(-2.6, -1.5)

Not saved

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